

CS5275 Problem Set 4

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Problem 1

We will (a) give an algorithm that uses exactly n queries, and show that it is (b) correct and (c) optimal.

- a. For each $i \in [n]$, query f at $x^{(i)}$ defined by

$$x_j^{(i)} = \begin{cases} 1, & j \neq i, \\ -1, & j = i, \end{cases}$$

i.e., its i^{th} coordinate is -1 and all other coordinates are 1. Set S to be the set of all values of i such that the query outputs -1 . Finally, output f with $f(x) = \chi_S(x)$.

- b. For any $i \in [n]$ and $S \subseteq [n]$, by definition we have

$$f(x^{(i)}) = \chi_S(x^{(i)}) = \prod_{j \in S} x_j^{(i)}.$$

By definition of $x^{(i)}$, if $i \notin S$, then every factor in the product is 1, so $f(x^{(i)}) = 1$, and if $i \in S$, then exactly one factor is -1 , so $f(x^{(i)}) = -1$. Thus by enumerating all $i \in [n]$, the algorithm correctly finds the set S and hence the function f .

- c. Note that f is determined by the set $S \subseteq [n]$ and there are 2^n subsets of $[n]$. As each query has exactly 2 possible outcomes, 1 or -1 , there are 2^k possible outcomes in total with k queries. For any algorithm to be correct in the worst case, it must be able to uniquely identify all 2^n possible f , so we must have $2^k \geq 2^n \implies k \geq n$, and thus at least n queries is required.

Problem 2

a. Consider the Fourier expansion of f as

$$f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S(x), \quad \chi_S(x) = (-1)^{\sum_{i \in S} x_i}.$$

By definition we have

$$\begin{aligned} \chi_S(x + y + z) &= (-1)^{\sum_{i \in S} (x+y+z)_i} = (-1)^{\sum_{i \in S} x_i + \sum_{i \in S} y_i + \sum_{i \in S} z_i} \\ &= (-1)^{\sum_{i \in S} x_i} (-1)^{\sum_{i \in S} y_i} (-1)^{\sum_{i \in S} z_i} = \chi_S(x) \chi_S(y) \chi_S(z). \end{aligned}$$

Thus we can write

$$\begin{aligned} f(x)f(y)f(z)f(x+y+z) &= \sum_{A \subseteq [n]} \hat{f}(A) \chi_A(x) \sum_{B \subseteq [n]} \hat{f}(B) \chi_B(y) \\ &\quad \sum_{C \subseteq [n]} \hat{f}(C) \chi_C(z) \sum_{D \subseteq [n]} \hat{f}(D) \chi_D(x+y+z) \\ &= \sum_{A, B, C, D \subseteq [n]} \hat{f}(A) \hat{f}(B) \hat{f}(C) \hat{f}(D) \chi_A(x) \chi_B(y) \chi_C(z) \chi_D(x+y+z), \end{aligned}$$

where

$$\begin{aligned} \chi_A(x) \chi_B(y) \chi_C(z) \chi_D(x+y+z) &= \chi_A(x) \chi_B(y) \chi_C(z) \chi_D(x) \chi_D(y) \chi_D(z) \\ &= (\chi_A(x) \chi_D(x)) (\chi_B(y) \chi_D(y)) (\chi_C(z) \chi_D(z)). \end{aligned}$$

Recall that the expectation

$$\mathbb{E}[\chi_S \chi_T] = \begin{cases} 1, & S = T, \\ 0, & S \neq T. \end{cases}$$

Therefore, by linearity of expectation,

$$\begin{aligned} \mathbb{E}[f(\mathbf{x})f(\mathbf{y})f(\mathbf{z})f(\mathbf{x}+\mathbf{y}+\mathbf{z})] &= \sum_{A, B, C, D \subseteq [n]} \hat{f}(A) \hat{f}(B) \hat{f}(C) \hat{f}(D) \\ &\quad \mathbb{E}[\chi_A(\mathbf{x}) \chi_D(\mathbf{x})] \mathbb{E}[\chi_B(\mathbf{y}) \chi_D(\mathbf{y})] \mathbb{E}[\chi_C(\mathbf{z}) \chi_D(\mathbf{z})]. \end{aligned}$$

All terms except those with $A = B = C = D$ vanish. Therefore

$$\mathbb{E}[f(\mathbf{x})f(\mathbf{y})f(\mathbf{z})f(\mathbf{x}+\mathbf{y}+\mathbf{z})] = \sum_{A=B=C=D \subseteq [n]} \hat{f}(A) \hat{f}(B) \hat{f}(C) \hat{f}(D) = \sum_{S \subseteq [n]} \hat{f}(S)^4,$$

as required.

b. Denote $p := \Pr_{\mathbf{x}, \mathbf{y}, \mathbf{z}} [f(\mathbf{x})f(\mathbf{y})f(\mathbf{z}) \neq f(\mathbf{x} + \mathbf{y} + \mathbf{z})]$ to be the required probability.

Note that

$$f(\mathbf{x})f(\mathbf{y})f(\mathbf{z})f(\mathbf{x} + \mathbf{y} + \mathbf{z}) = \begin{cases} 1, & f(\mathbf{x})f(\mathbf{y})f(\mathbf{z}) = f(\mathbf{x} + \mathbf{y} + \mathbf{z}), \\ -1, & f(\mathbf{x})f(\mathbf{y})f(\mathbf{z}) \neq f(\mathbf{x} + \mathbf{y} + \mathbf{z}). \end{cases}$$

Thus $\mathbb{E} [f(\mathbf{x})f(\mathbf{y})f(\mathbf{z})f(\mathbf{x} + \mathbf{y} + \mathbf{z})] = 1(1 - p) + (-1)p = 1 - 2p$.

By part (a), we have

$$p = \frac{1}{2} \left(1 - \sum_{S \subseteq [n]} \hat{f}(S)^4 \right),$$

so to lower bound p , we just need to upper bound $\sum_{S \subseteq [n]} \hat{f}(S)^4$.

Recall the identity $\hat{f}(S) = 1 - 2 \text{dist}(f, \chi_S)$. As f is ε -far from every parity function and its negation, by definition we have $|\hat{f}(S)| \leq 1 - 2\varepsilon$ for every $S \subseteq [n]$.

Denote $a_S := \hat{f}(S)^2$. We have $\max_{S \subseteq [n]} a_S \leq (1 - 2\varepsilon)^2$. Also recall the corollary of Parseval's theorem, $\sum_{S \subseteq [n]} a_S = 1$.

As $a_S \geq 0$, we have

$$a_S^2 \leq a_S \max_{S \subseteq [n]} a_S \implies \sum_{S \subseteq [n]} a_S^2 \leq \left(\sum_{S \subseteq [n]} a_S \right) \left(\max_{S \subseteq [n]} a_S \right) \leq (1 - 2\varepsilon)^2.$$

Hence

$$p = \frac{1}{2} \left(1 - \sum_{S \subseteq [n]} a_S^2 \right) \geq \frac{1}{2} (1 - (1 - 2\varepsilon)^2) = 2\varepsilon(1 - \varepsilon).$$

Note that for f to be ε -far, by definition,

$$\Pr_{\mathbf{x}} [f(\mathbf{x}) \neq \chi_S(\mathbf{x})] \geq \varepsilon \text{ and } \Pr_{\mathbf{x}} [f(\mathbf{x}) \neq -\chi_S(\mathbf{x})] \geq \varepsilon.$$

But for any x , either $f(x) = \chi_S(x)$ or $f(x) = -\chi_S(x)$, so

$$\Pr_{\mathbf{x}} [f(\mathbf{x}) \neq \chi_S(\mathbf{x})] + \Pr_{\mathbf{x}} [f(\mathbf{x}) \neq -\chi_S(\mathbf{x})] = 1 \implies \varepsilon \leq \frac{1}{2}.$$

Thus $p \geq 2\varepsilon(1 - \varepsilon) \geq \varepsilon \in \Omega(\varepsilon)$ as required.

Problem 3

Recall the following definitions: for all $x, y \in \{-1, 1\}^n$,

- f is monotone: $f(x) \leq f(y)$ whenever $x_i \leq y_i$ for all i ,
- f is odd: $f(-x) = -f(x)$,
- f is symmetric: $f(x^\pi) = f(x)$ for all permutations $\pi \in S_n$.

(i) We first prove that n is odd. Suppose otherwise. Then we can construct $x = (1, \dots, 1, -1, \dots, -1)$ with an equal number of 1's and -1 's. As f is odd, we have $f(-x) = -f(x)$. But note that $-x = (-1, \dots, -1, 1, \dots, 1)$ is a permutation of x . As f is symmetric, we have $f(-x) = f(x)$, so $f(x) = 0$. But the range of f is $\{-1, 1\}$, so contradiction.

(ii) Now we show that $f = \text{Maj}_n$, the majority function.

As f is symmetric, its value only depends on the number of 1's in the input. In other words, there exists a function $g : \{0, 1, \dots, n\} \rightarrow \{-1, 1\}$ such that $f(x) = g(k)$ for $k = |\{i : x_i = 1\}|$.

As f is odd, $f(-x) = -f(x)$, and if x has k 1's, then $-x$ has $n - k$ 1's. Hence $g(n - k) = -g(k)$. As n is odd, take $k = \frac{n-1}{2}$, so that $g\left(\frac{n+1}{2}\right) = -g\left(\frac{n-1}{2}\right)$.

As f is monotone, flipping -1 's to 1's cannot decrease its value. Thus g is also non-decreasing. Hence $g\left(\frac{n-1}{2}\right) \leq g\left(\frac{n+1}{2}\right)$.

As the range of g is $\{-1, 1\}$, we must have $g\left(\frac{n-1}{2}\right) = -1$ and $g\left(\frac{n+1}{2}\right) = 1$. But g is non-decreasing, which implies

$$g(k) = \begin{cases} -1, & k \leq \frac{n-1}{2}, \\ 1, & k \geq \frac{n+1}{2}. \end{cases}$$

Therefore by definition

$$f(x) = \begin{cases} -1, & |\{i : x_i = 1\}| \leq \frac{n-1}{2}, \\ 1, & |\{i : x_i = 1\}| \geq \frac{n+1}{2}. \end{cases}$$

In the first case, there are more -1 's than 1's, so $x_1 + x_2 + \dots + x_n < 0$ and $\text{sgn}(x_1 + x_2 + \dots + x_n) = -1$. In the second case, there are more 1's than -1 's, so $x_1 + x_2 + \dots + x_n > 0$ and $\text{sgn}(x_1 + x_2 + \dots + x_n) = 1$.

Hence we have $f(x) = \text{sgn}(x_1 + x_2 + \dots + x_n) \implies f = \text{Maj}_n$ as required.

Problem 4

Consider the d -dimensional hypercube graph $G = (V, E)$ where $V = \{-1, 1\}^d$, $|V| = 2^d$ and $|E| = d2^{d-1}$.

Recall that the conductance is defined as

$$\Phi(G) = \min_{S \subseteq V: S \neq V \wedge S \neq \emptyset} \frac{|E(S, V \setminus S)|}{\min(\text{vol}(S), \text{vol}(V \setminus S))}.$$

WLOG assume that $0 < |S| \leq |V|/2 = 2^{d-1}$. As G is d -regular, $\text{vol}(S) = d|S|$, and

$$\Phi(G) = \min_{S \subseteq V: 0 < |S| \leq 2^{d-1}} \frac{|E(S, V \setminus S)|}{d|S|}.$$

To prove that $\Phi(G) = 1/d$, we show that (a) $\Phi(G) \geq 1/d$, (b) $\Phi(G) \leq 1/d$.

a. For any $S \subseteq V$ with $0 < |S| \leq 2^{d-1}$, consider its (Boolean) indicator function f ,

$$f(x) = \begin{cases} 1, & x \in S, \\ -1, & x \notin S. \end{cases}$$

Then we have

$$\Pr[f(\mathbf{x}) = 1] = \frac{|S|}{|V|} = \frac{|S|}{2^d}, \quad \Pr[f(\mathbf{x}) = -1] = \frac{|V \setminus S|}{|V|} = \frac{2^d - |S|}{2^d}.$$

Therefore its expectation

$$\mathbb{E}[f] = 1 \cdot \Pr[f(\mathbf{x}) = 1] + (-1) \cdot \Pr[f(\mathbf{x}) = -1] = \frac{|S|}{2^d} - \frac{2^d - |S|}{2^d} = \frac{|S|}{2^{d-1}} - 1.$$

As the range of f is $\{-1, 1\}$, $f(x)^2 = 1 \implies \mathbb{E}[f^2] = 1$. Therefore its variance

$$\text{Var}[f] = \mathbb{E}[f^2] - \mathbb{E}[f]^2 = 1 - \left(\frac{|S|}{2^{d-1}} - 1 \right)^2 = \frac{2|S|}{2^{d-1}} - \frac{|S|^2}{2^{2(d-1)}} = \frac{|S|(2^d - |S|)}{2^{2(d-1)}}.$$

As $0 < |S| \leq 2^{d-1}$, we have $|S|(2^d - |S|) \geq 2^{d-1}|S|$, so

$$\text{Var}[f] \geq \frac{2^{d-1}|S|}{2^{2(d-1)}} = \frac{|S|}{2^{d-1}}.$$

From Poincaré inequality, we have $\text{Var}[f] \leq \text{I}[f]$, so $\text{I}[f] \geq \frac{|S|}{2^{d-1}}$.

Recall that $\frac{\text{I}[f]}{d}$ is the fraction of the edges that cross the cut represented by f in the d -dimensional hypercube. In this case, this means

$$\frac{\text{I}[f]}{d} = \frac{|E(S, V \setminus S)|}{|E|} = \frac{|E(S, V \setminus S)|}{d2^{d-1}} \implies |E(S, V \setminus S)| = 2^{d-1} \text{I}[f] \geq |S|.$$

Hence the conductance

$$\Phi(G) = \min_{S \subseteq V: 0 < |S| \leq 2^{d-1}} \frac{|E(S, V \setminus S)|}{d|S|} \geq \frac{|S|}{d|S|} = \frac{1}{d}.$$

- b.** Take the set $S = \{x \in \{-1, 1\}^d : x_1 = 1\}$, i.e., where the first coordinate is 1. Then $|S| = 2^{d-1}$. Note that any edges between S and $V \setminus S$ must have their first coordinate flipped. There are 2^{d-1} such edges as all the other $d - 1$ coordinates are free. Hence

$$\Phi(G) \leq \frac{|E(S, V \setminus S)|}{d|S|} = \frac{2^{d-1}}{d2^{d-1}} = \frac{1}{d}.$$

From parts (a) and (b), we conclude that $\Phi(G) = 1/d$ as required.